Simply Rugby System

Inception Phase Planning

Contents

1. Project Overview

Simply Rugby, a Scottish rugby club, has commissioned the development of a new computerised system to monitor player progress. The Inception Plan includes an analysis of key information such as project objectives, requirements, and an initial project plan.

• Purpose of the System

The Simply Rugby system is designed to provide an efficient way for coaches to track player details for each squad, skill development, game information and training session records. This section overviews the system's purpose, scope, and background of Simply Rugby Club.

- ### Scope of the Project

- ❖ Managing profiles – It includes creating, deleting, and editing profiles, such as the ability to change an account from Staff to Coach or from Coach to Staff if a person is no longer a coach but remains in the club.
- ❖ Managing player details – stores detailed information about players, including name, age and statistics.
- ❖ Managing team rosters - creating, editing, and assigning players to different teams.
- ❖ Tracking player skill development - monitors players progress based on training sessions and match performance.
- ❖ Recording game details - system stores information about played matches.
- ❖ Managing training sessions - keeps track of training plans, player attendance, and performance.
- ❖ Storing details of all club members – persistent data maintains records for coaches, managers and staff.
- ❖ The prototype is designed to support multiple teams, with an initial setup of three teams. However, the number of teams is not fixed, and the system will allow administrators to add new teams as needed.
- ❖ The Fixture Secretary will be able to organize matches.
- ❖ The Membership Secretary will have access to player data and training information.

- ### Overview of Simply Rugby Club

Simply Rugby, supported by Scottish Rugby Union (SRU) operates multiple squads playing at all of the age grades across mini, midi and senior levels, with coaches who are normally parents of players managing players development and performance.

2. Analysis of the Project Brief

This Section shows client expectations and the essential system functionalities.

- ### Sources of Information - Conversation with Chairman

A functioning login system is required to control entry to the system individually.

The details of who needs to log in. There are different tiers of permissions. GUI design should be professional-looking and easy to use.

For purposes of the prototype, only a Windows PC application is required.

Data in the system should be persistent.

Logged-in users should be able to edit data and save their changes.

Compliance with data protection legislation and insurance, health and safety requirements.

Include user documentation explaining how to use it.

The one-to-one relationship between coach and squad. Anyone can apply to the club membership secretary to join. Disclosure and Barring Service (DBS) status.

- ### Project Constraints and Considerations

The project must be completed within 3 months. The prototype will be built as a desktop application using local storage with no external access (no web or mobile version). Secure data by allowing access only to logged-in users and ensure legal compliance by properly storing data. Access is restricted to administrators and coaches.

3. Initial Functional and Non-Functional Requirements

Functional requirements define what the product should do, specifying the features it should offer to users, but non-functional requirements describe how the system should operate, ensuring aspects such as performance parameters, security, and quality.

- ## Functional Requirements

All authorized users must be able to log in and log out securely.

The system must differentiate between user roles (Coach, Administrator, Club Member) and have permissions accordingly.

Unauthorized users should not have access to the system or its data.

Functionalities based on role in a team:

Coach	Administrator	Club Member – read-only	Membership Secretary	Fixture Secretary
Can create, edit, and delete player profiles	Has full control over system settings. Can make Coach Profile	Can view assigned team information	Training session details	Organising fixtures
Can assign players to teams and update team rosters	Can create, edit, and remove user accounts	Can check training schedules	Show player details	
Can mark attendance	Can modify team structures and club-wide settings	Can check players details		
Can record and update player performance statistics	Can access all players and statistics	Cannot modify or edit any data		
Can view team and player details				

The system must record and display game results and individual player statistics.

Coaches and administrators should be able to schedule training sessions.

Attendance tracking should be available for training sessions.

All user data must be securely stored and protected.

The system must comply with data protection regulations.

Only logged-in users with the appropriate permissions should have access to sensitive data.

- ## Non-Functional Requirements

The system should respond to user actions under normal load conditions (turn on).

The prototype should handle a minimum of 1 concurrent user without performance degradation.

Users must use a secure login system (password hashing and role-based access).

Only authorized users should access or modify system data.

The prototype must follow GDPR (General Data Protection Regulation) and DBS compliance.

The user interface (UI) should be intuitive and easy to navigate.

The system should be accessible on a Windows-based desktop environment.

Data should be automatically saved after every major action like exit application.

4. Information Gathering and Background Research

- ## Research about Similar Systems

To create a simple rugby system, it is worth gathering information from existing solutions such as SportMember and LoveAdmin, which offer member management, attendance tracking, and various other features to gain a better understanding of the project.

- ## Best Practices in Sports Club Management Systems

The best practices I have come across in sports club management systems emphasise simplicity. Clear user role management ensures that access to data is restricted to the designated roles. An intuitive, minimalist interface enhances user-friendliness, making it easier for older users to navigate the application.

- ## Compliance with Data Protection and Legal Considerations

  - General Data Protection Regulation (GDPR) 2016 - protection of natural persons with regard to the processing of personal data and on the free movement of such data
  - Data Protection Act 2018 - prevent unauthorized access, loss, or misuse
  - Children's Act 1989 & 2004 (UK) - safeguarding of children
  - Disclosure and Barring Service (DBS) - criminal record checks
  - Health and Safety at Work Act 1974 (UK) - safe working conditions

5. Aims of the Project

- ## Short-Term Plans

Actions that have already been taken in this project, as well as those that will be implemented until the final version is completed. This version will be released as a prototype and sent to Simply Rugby Club for evaluation.

Stage 1: Planning

Further analysis of the entire project in terms of functional and non-functional requirements.

Prepare an initial use case model, sequence diagram and class diagram for system functionalities.

Gather additional information from the client.

Make a project plan with milestones, tasks, and timelines.

Prepare initial system documentation.

Stage 2: development

Everything that is included in the Functional Requirements section.

Unit and system testing to confirm that the functions operate as intended.

Confirm that all data operations CRUDE (Create, Read, Update, Delete) are properly working.

Make (GUI) a graphical user interface for better usability and access.

Stage 3: Evaluation

Check that the system meets both its functional and non-functional requirements.

Provide full end user and technical documentation.

- ### Long-Term Vision

Features that may be added after approval. These are potential enhancements that could be implemented as the application evolves and expands.

- ✓ Add extra functionality to support mobile or web access within the timeframe
- ✓ Database solution that we will have to integrate into our application
- ✓ Separate system to send data to players
- ✓ Scottish Rugby SCRUMS connected to architecture
- ✓ Functionality to players
- ✓ Move to cloud storage if the system will grow fast
- ✓ Generate reports

6. Resources and Materials Required

- ### Development Environment (Tools & Software)

Programming Language	Java
Integrated development environment (IDE)	Eclipse
Database	None - local
Version Control	Github
UML Modeling Tools	Lucidchart
Testing Framework	JUnit
GUI Framework	Swing
Project Management	MS Projects

- ### Hardware Requirements

Windows-based PC with a minimum of Intel i5 gene 11 plus, 256GB SSD and 8GB RAM.

- ### Personnel and Expertise Needed

The person responsible for the Software Developer, Project Manager, and UI Designer roles, covering programming, debugging, implementation, project management, and user interface design, will be Kamil Milej.

Client/Stakeholder (Simply Rugby Chairman) - Providing feedback and validation throughout the project.

Testing (test plan and test runs) also Kamil Milej

7. Key Sources of Information

Key materials that contributed to the development of this Inception Phase Planning:

- Initial brief provided by Simply Rugby
- Additional information document provided by Simply Rugby
- Scottish Rugby Union (SRU) Guidelines – The official website helped understand the topic and will continue supporting the project's implementation.
- Research on Similar Systems – Applications such as SportMember and LoveAdmin contributed to the analysis of this system.
- Data Protection and Legal Regulations
- Additional information provided during conversations with the Chairman

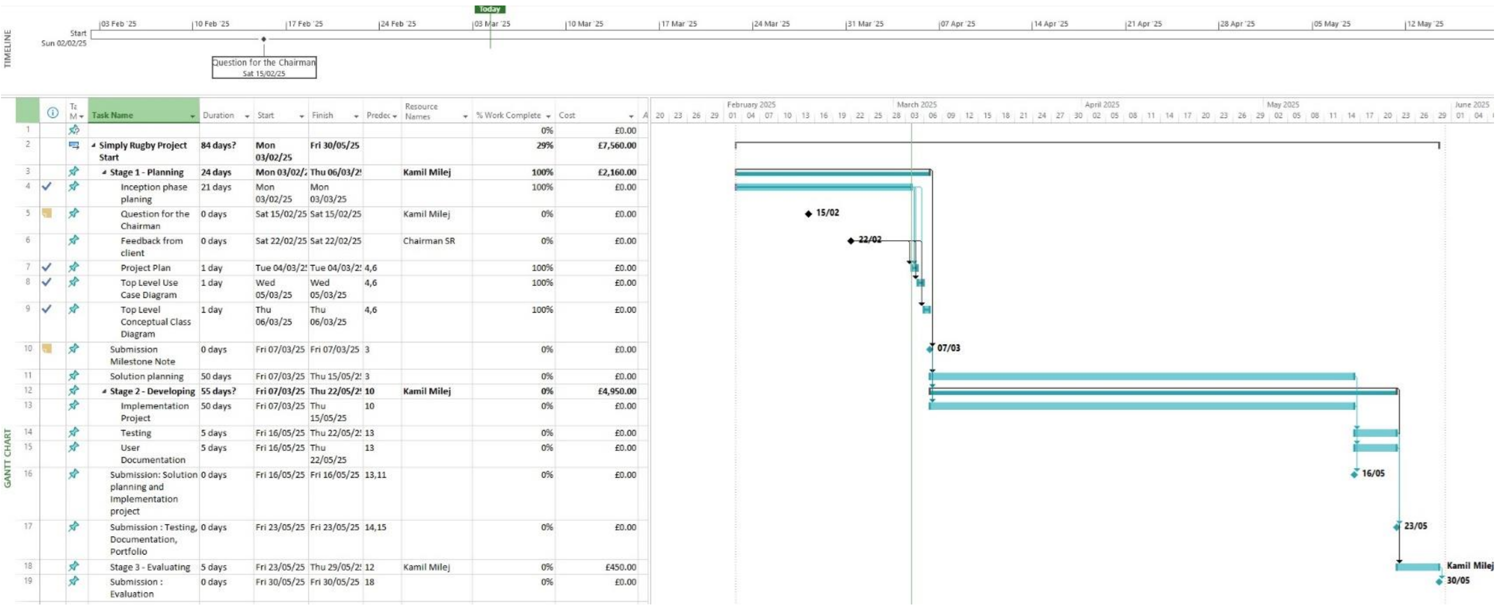
AltexSoft. (2021). *Software Planning and Technical Documentation*. [Online video]. Available at https://www.youtube.com/watch?v=2qlcY9LkFik [Accessed 2 March 2025]

Imagine Projects. (2021). *Project Inception Phase*. [Online]. Available at https://imagineprojects.co.uk/project-inception-phase/ [Accessed 3 March 2025]

8. Project Plan

The project plan has been created for better project management and can be accessed by Simply Rugby at the following link:
https://drive.google.com/file/d/1QxSbg_hTLac5V8ZSJO0rUL8woH4SyFvv/view?usp=sharing

8.1. General Overview



A brief description of what this plan is all about:

The Simply Rugby project is divided into three stages. Each stage usually ends with a milestone that marks the completion of the stage but is not included within it, except for the first milestone, which occurred during Stage 1.

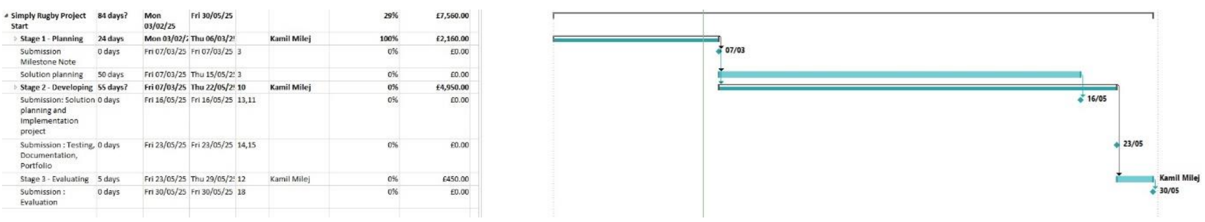
Each stage has its own subtasks related to that specific stage. Task dependencies are based on two factors:

Time-based dependencies – when one task finishes, the next one starts.

Logical dependencies – when a task cannot start until a necessary prerequisite is completed.

For example, Testing cannot begin without the completion of the Implementation Project.

8.2. Shorter version



8.3. Resources

Effective Date	Standard Rate	Overtime Rate	Per Use Cost
--	£450.00/w	£0.00/h	£0.00

The plan includes two types of resources:

- Added workers – individuals who contributed to the plan.
- Work-related costs – in our case, an employee earns £450 per week, as the initial assumption is £450 per week.

This amount may change after further discussions with the chairman of Simply Rugby.

9. Conclusion

The Project Plan was created by the person working on the project to outline how the system will develop. The document is divided into sections that contain key information.

The document will be updated as needed based on project progress.